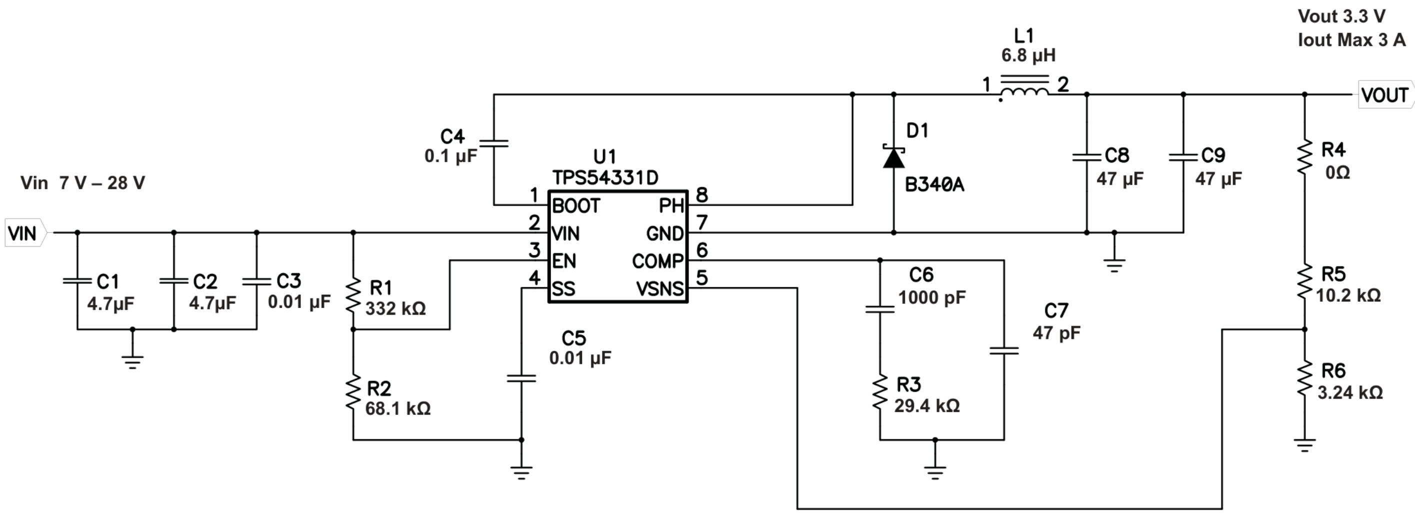
TPS54331 Buck — 12 V in → 5 V / 3 A and 4 V / 3 A — Cheat Sheet

Fixed **12 V** input • fixed **570 kHz** • V_{REF} = 0.800 V • current-mode, Type-II compensation. **One part per position** — every LCSC number below was checked on its own lcsc.com page (stock ≥ 1000) on 2026-07-14. Alternates and the upgrade path live in the companion **TPS54331_5V_4V_BOM.xlsx**.



*Reference topology only (TI SLVS839B, Figure 13). The values printed on the figure are the **original 3.3 V design** — use the tables below instead.*

Design equations

$V_{OUT} = 0.8 \times (1 + R5/R6)$

$L_{MIN} = V_O(V_{IN} - V_O) / (V_{IN} \cdot K_{IND} \cdot I_O \cdot f_{SW})$ $[K_{IND} = 0.3$ for ceramics$]$

$\Delta I_L = V_O(V_{IN} - V_O) / (V_{IN} \cdot L \cdot f_{SW})$ • $I_{PK} = I_O + \Delta I_L/2$ • $I_{RMS} = \sqrt{(I_O^2 + \Delta I_L^2/12)}$

$R_Z = 2\pi \cdot f_{CO} \cdot V_O \cdot C_O \cdot R_{OA} / (gm_{COMP} \cdot V_{ggm} \cdot V_{REF})$ $[R_{OA}=8\ M\Omega,\ gm=12\ A/V,\ V_{ggm}=800,\ f_{CO} \le 25\ kHz]$

$C_Z = 1/(2\pi\ f_Z\ R_Z)$ • $C_P = 1/(2\pi\ f_P\ R_Z)$ $[f_Z \approx 0.68\ kHz,\ f_P \approx 82\ kHz]$ • $C_{SS} = T_{SS} \cdot I_{SS} / V_{REF}$

$R1 = (V_{START} - V_{STOP})/3\ \mu A$ • $R2 = 1.25 / [\ (V_{START} - 1.25)/R1 + 1\ \mu A\ ]$

BOM A — 5 V @ 3 A (divider gives exactly 5.000 V • duty 45 % • $1.49/board at qty 100)

Ref	Qty	Value	Pkg	LCSC	Manufacturer / MPN	Stock	$ @100	Function
U1	1	TPS54331DR	SOIC-8	C9865	TI TPS54331DR	48,460	$0.3594	570 kHz buck controller
L1	1	10 µH	11.5x10 mm	C475915	cjiang FXL1040-100-M	85,795	$0.2866	8.5 A sat / 7.8 A rms, 30 mOhm
C8, C9	2	22 µF 16 V X7R	1210	C55530	Samsung CL32B226KOJNNNE	4,240	$0.1710	output filter -- sets the loop
C1, C2	2	4.7 µF 50 V X7R	1206	C29823	FH 1206B475K500NT	84,810	$0.1757	input bulk; I_rms = 1.5 A
C3	1	0.01 µF 50 V X7R	0603	C1589	Samsung CL10B103KB8NNNC	528,750	$0.0241	input HF bypass
C4	1	0.1 µF 50 V X7R	0603	C1591	Samsung CL10B104KB8NNNC	1,691,500	$0.0233	BOOT (fixed value)
C5	1	0.01 µF 50 V X7R	0603	C1589	Samsung CL10B103KB8NNNC	528,750	$0.0241	slow start -> 4 ms
D1	1	SS34 40 V / 3 A	SMA	C7420365	R+O SS34	838,880	$0.0355	catch diode; hottest part on the board
R1	1	332 kΩ 1%	0603	C23139	UNI-ROYAL 0603WAF3323T5E	231,800	$0.0033	EN / UVLO
R2	1	68.1 kΩ 1%	0603	C25976	UNI-ROYAL 0603WAF6812T5E	28,600	$0.0029	start 7.0 V / stop 6.0 V
R4	1	0 Ω	0603	C95177	YAGEO RC0603JR-070RL	997,200	$0.0047	loop break for stability probing
R5	1	10.5 kΩ 1%	0603	C2933127	FOJAN FRC0603F1052TS	14,200	$0.0040	FB top
R6	1	2.00 kΩ 1%	0603	C105576	YAGEO RC0603FR-072KL	3,300	$0.0066	FB bottom -> 5.000 V
R3	1	28.0 kΩ 1%	0603	C22989	UNI-ROYAL 0603WAF2802T5E	142,000	$0.0029	Rz
C6	1	8200 pF 50 V X7R	0603	C1322360	FH 0603B822K500NT	79,100	$0.0110	Cz
C7	1	68 pF 50 V C0G	0603	C1680	FH 0603CG680J500NT	75,900	$0.0114	Cp

BOM B — 4 V @ 3 A (divider gives exactly 4.000 V • duty 37 % • $1.52/board at qty 100)

Ref	Qty	Value	Pkg	LCSC	Manufacturer / MPN	Stock	$ @100	Function
U1	1	TPS54331DR	SOIC-8	C9865	TI TPS54331DR	48,460	$0.3594	570 kHz buck controller
L1	1	10 µH	11.5x10 mm	C475915	cjiang FXL1040-100-M	85,795	$0.2866	8.5 A sat / 7.8 A rms, 30 mOhm
C8, C9	2	22 µF 16 V X7R	1210	C55530	Samsung CL32B226KOJNNNE	4,240	$0.1710	output filter -- sets the loop
C1, C2	2	4.7 µF 50 V X7R	1206	C29823	FH 1206B475K500NT	84,810	$0.1757	input bulk; I_rms = 1.5 A

Ref	Qty	Value	Pkg	LCSC	Manufacturer / MPN	Stock	$ @100	Function
C3	1	0.01 µF 50 V X7R	0603	C1589	Samsung CL10B103KB8NNNC	528,750	$0.0241	input HF bypass
C4	1	0.1 µF 50 V X7R	0603	C1591	Samsung CL10B104KB8NNNC	1,691,500	$0.0233	BOOT (fixed value)
C5	1	0.01 µF 50 V X7R	0603	C1589	Samsung CL10B103KB8NNNC	528,750	$0.0241	slow start -> 4 ms
D1	1	SS34 40 V / 3 A	SMA	C7420365	R+O SS34	838,880	$0.0355	catch diode; hottest part on the board
R1	1	332 kΩ 1%	0603	C23139	UNI-ROYAL 0603WAF3323T5E	231,800	$0.0033	EN / UVLO
R2	1	68.1 kΩ 1%	0603	C25976	UNI-ROYAL 0603WAF6812T5E	28,600	$0.0029	start 7.0 V / stop 6.0 V
R4	1	0 Ω	0603	C95177	YAGEO RC0603JR-070RL	997,200	$0.0047	loop break for stability probing
R5	1	10.2 kΩ 1%	0603	C22772	UNI-ROYAL 0603WAF1022T5E	92,600	$0.0027	FB top
R6	1	2.55 kΩ 1%	0603	C2933178	FOJAN FRC0603F2551TS	26,900	$0.0026	FB bottom -> 4.000 V
R3	1	23.7 kΩ 1%	0603	C22912	UNI-ROYAL 0603WAF2372T5E	434,800	$0.0030	Rz
C6	1	10 nF 50 V X7R	0603	C1589	Samsung CL10B103KB8NNNC	528,750	$0.0241	Cz (same part as C3/C5)
C7	1	82 pF 50 V C0G	0603	C1683	FH 0603CG820J500NT	74,100	$0.0285	Cp

Why 22 µF on the output (and not 10 µF)

C8/C9	Effective @ 5 V	Ripple	Droop on a 1.5 A step	Verdict
2 × 22 µF 16 V	34 µF	8 mV	281 mV	**this build** — best value
2 × 47 µF 16 V	54 µF	5 mV	177 mV	upgrade: +$0.20/bd, needs new R3/C6/C7 (see XLSX)
2 × 10 µF 16 V	13 µF	18 mV	**735 mV**	rejected — saves 10c, sags 15%

Ceramics lose most of their value under DC bias, so 2 × 10 µF/16 V is only ~13 µF at 5 V. The ripple still passes — it's the **transient** that fails. **C8/C9 are part of the feedback loop**, so you can never swap output caps without also changing R3/C6/C7.

Inductor — the one real design decision

	5 V / 3 A	4 V / 3 A	vs. the 3.5 A minimum current limit
10 µH — **this build**	ΔI_L 0.51 A I_{PK} **3.26 A**	ΔI_L 0.47 A I_{PK} **3.23 A**	0.24 A of margin — comfortable
6.8 µH — TI's value	ΔI_L 0.75 A I_{PK} **3.38 A**	ΔI_L 0.69 A I_{PK} **3.34 A**	0.12 A of margin — **TIGHT, no margin**

The TPS54331's **guaranteed-minimum current limit is 3.5 A** (typ 5.8 A). At 3 A load the 6.8 µH inductor from TI's typical-app circuit peaks at 3.38 A — legal, but a worst-case part leaves you 0.12 A. **10 µH doubles the margin** for the same compensation and lower ripple.

Operating numbers (computed, L = 10 µH)

	5 V / 3 A	4 V / 3 A
Duty cycle	45 %	37 %
Inductor I_{RMS}	3.00 A	3.00 A
Output ripple	~8 mV pp	~7 mV pp
IC dissipation (typ / max $R_{DS(on)}$)	0.44 W / 0.70 W	0.38 W / 0.59 W
Catch-diode dissipation	0.83 W	0.95 W
L_{MIN} (K_{IND} = 0.3)	5.7 µH	5.2 µH

Checks & gotchas

1. Your 6.3 V output caps must go. C8/C9 in the 3.3 V design are 47 µF / **6.3 V** — at 5 V out that fails on voltage rating *and* on DC-bias derating. The 16 V parts above replace them.
2. 3 A is the edge of this chip. Use 10 µH — see the inductor table.
3. Thermals. SOIC-8 θ_{JA} is **100 °C/W**, so the IC's 0.44–0.70 W is a **44–70 °C rise** over ambient — you need the datasheet's ground-plane copper under the GND pin. **D1 dissipates more than the IC** (~0.95 W at 4 V): it is the hottest part.
4. LCSC stock ≠ JLCPCB stock — check per part. A part can show deep stock in JLCPCB's library and still be dead on lcsc.com. Every SAMWHA `CS3225...` cap is; so are *some* UNI-ROYAL `0603WAF...` resistors (1.91 k, 10.0 k) while others (332 k, 28.0 k) are deeply stocked. Don't assume by brand.
5. Why not TI's 10.0 k / 1.91 k divider? 1.91 kΩ is unbuyable in 0603 on LCSC (every brand OOS). 10.5 k / 2.00 k hits **5.000 V exactly** — better than TI's 4.99 V.
6. EN pin. Since $V_{IN} > V_{OUT}$ + 2 V the UVLO divider is optional — EN may float. Keep 332 k / 68.1 k, or fit 51.1 k (C23072) for R2 to start at 9 V on a 12 V rail.

Only R5, R6, R3, C6 and C7 differ between the two rails — U1, L1, C8, C9 and everything else is shared. Board cost at qty 100: **$1.49** (5 V) / **$1.52** (4 V), excl. PCB. Prices are LCSC list, 2026-07-14, and move — re-check before ordering.